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ENHANCED PYTHONETICS IMPLEMENTATION

This module provides an enhanced implementation of the Pythonetics framework, integrating the advanced sovereign equation with external verification components.

Third-order evolution of cybernetics with the TrueAlphaSpiral framework, combining factual verification, ethical analysis, and configuration management.

Architect: Russell Nordland

Enhanced by: TrueAlphaSpiral Team

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ConfigManager)?FactualVerifier)?EthicalAnalyzerz'%(asctime)s [%](levelname)s] %
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Orchestrates verification using the Universal Laws integrated with TrueAlphaSpiral principles.

Args:

- text: The text to verify
- verify_as: Type of verification (claim, wisdom, pattern, etc.)

Returns:

Dict containing verification results

N

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corerecursion_depthdimensionalAlignmentrhythmPatternspolarityAnalysisr0  
causalChains  
truthScore)  
truth_factordistance_factor  
size_factorsovereigntyScorer?r?drr?success)?status?    timestamp?  
text_hash?analysis)?hashlib?md5?encode?  
hexdigestr"?reload_if_changedr(apply_correspondence?_analyze_dimensional_alignment?  
_apply_rhythm?_apply_polarity?_trace_causality?_apply_integration?update?_calculate_sov-  
ereignty?_calculate_distance_factor?_calculate_size_factor?  
_log_audit?_adjust_weightstr&?append?len?pop?_calculate_truth_alignmentr%)  
r,r1r2rGr7?result?rhythm_patterns?polarity_analysis?hybrid_results?sovereignty_scores  
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Recursive truth validation across multiple dimensions.

Args:

text: The text to verify

verify_as: Type of verification

depth: Current recursion depth

Returns:

Dict containing verification results

rrz-Insufficient recursion depth for

verification)r=?factualConfidence?truthResonance?consistencyScore?selfReferenceIndex?de-
ceptionPatterns?suggestedActionsrrr r!333333?rBz#Deepening truth recursion at depth
?ffffff?r=rfrgrh?)

?_calculate_factual_score?_calculate_truth_resonance?_calculate_consistency?_calculate_-
self_referencer)?minr*r+rN?_identify_deception_patterns?_generate_suggested_actions?rroun
d)

r,r1r2rd?

factual_score?truth_resonance?consistency_score?self_reference?

truth_score?dimension_scores?deeper_results?deception_patterns?suggested_actionsss

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Analyzes text alignment across multiple dimensions.

Args:
text: The text to analyze

Returns:
List of dimensional alignment results

```
r6?truth_dimensionsrrzFactual DomainrzConceptual Domainr zEthical Domainr!
zPhenomenological Domainrz DomainzComplete Disharmony暂??zStable Alignment??zPartial
Harmonyg??zSubtle DissonancerzSignificant Misalignmentrn)? dimension?
alignment?resonanceState) r"r(ro?_calculate_conceptual_score?_calculate_ethical_score?!
_calculate_phenomenological_score?titlerZrv)r,r1?
dimensions?dimension_results?dim?score?name?states r-
rOz2EnhancedPythonetics._analyze_dimensional_alignment?sz???(?,?,?V?5G?
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Identifies cyclical patterns in truth verification.

Args:
text: The text to analyze

Returns:
List of rhythm patterns or None

```
r6?rhythm_cycle_length??rhythm_analysis?similarity_thresholdrm?minimum_logs_requiredr8
?improvement_thresholdg???decline_thresholdg??Ni?r1rFr^r=rFr=?
similarityc??|dS)NrF?)?xs r-
?<lambda>z3EnhancedPythonetics._apply_rhythm.<locals>.<lambda>@s
??Q?{(?^?r/) ?keyc??g[]
}d??
S)r=r?)?.0?vs r-?
<listcomp>z5EnhancedPythonetics._apply_rhythm.<locals>.<listcomp>Cs???E?E?!?a?
?o?E?E?Er/c3?B?K?[]}?|?|dzkV???dS?rBNr???r???i?scoress ?r-?
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<genexpr>z4EnhancedPythonetics._apply_rhythm.<locals>.<genexpr>Es4??Q??Q??v?a
?y?6?!?a?%?=0?Q??Q??Q??Q??Qr/rBzAscending
Truth?333333??)?pattern?strengthc3?B?K?[]}?|?|dzkV??dSr?r?s ?r-
r?z4EnhancedPythonetics._apply_rhythm.<locals>.<genexpr>Gs4??S?S?q???V?A
??E?]?2?S?S?S?S?S?Sr/zDescending Truth?rzNet-Positive Evolutionr??c3?2K?[]}|
ddV??dS?r^r=Nr??r??logs r-
r?z4EnhancedPythonetics._apply_rhythm.<locals>.<genexpr>Ss+???O?O?C?C??M?,?
7?O?O?O?O?O?O?Or/c3?2K?[]}|ddV??dSr?r?s r-
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8?Q?Q?Q?Q?Q?Q?Qr/zCyclical Improvementg??zCyclical Decline)
r"r(r[r'?_calculate_text_similarityrZ?sort?all?range?sum)r,r1?
cycle_lengthr??
minimum_logsr?r??similar_verificationsr?r??
recent_logs?
first_cycle?
second_cycle? first_avg?
second_avgr?s @r-rPz!EnhancedPythonetics._apply_rhythms-
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Analyzes opposing truth forces within text.

Args:
text: The text to analyze

Returns:
Dict with polarity analysis
nlp?polarity?opposition_markers)?but?however?yet?although?
nonetheless?opposition_density_thresholdg??rrB?
?UnifiedrzStrong OppositionrzModerate OppositionzSubtle Oppositionrnr?)?
polarityState?oppositionDensity?synthesisScore)r"r(?splitr[r??lower?maxrv)
r,r1r??opposition_threshold?words?
text_length?

oppositionsr??opposition_density?polarity_states
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Traces causal chains within text.

Args:
text: The text to analyze

Returns:
List of causal chains

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r? causality?markers)?because? therefore?since?so?thuszleads
to?causes?implied_causality_threshold?333333??rrBrm?ffffff?)?marker?positionr?r?暫
??r??impliedr?)r"r(r?r?? enumerater[rZ?random?uniform)
r,r1?causal_markersr?r??
causal_chainsr??word?implied_causalitys r-
rRz$EnhancedPythonetics._trace_causality?s}????,?0?0?
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Balances structured logic with intuitive learning.

Args:

results: Current verification results

Returns:

Dict with balanced verification results

rfr?rBg?MbP?zPerfect Balanceg?zLogic Dominantgq=

rp??zIntuition Dominantrn)?

balanceRatio?

balanceState)?absr?rv)r,r??

logical_score?

ideal_logical?intuition_score?ratio?

balance_states r-rSz&EnhancedPythonetics._apply_integration?s?? 3?4?
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s?==#@?A?A?M?Q?R??E?? ; ; ;?*?
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a
 Logs AI's perception of its verification process (Self-Awareness Law).

Args:
 text: The verified text
 text_hash: Hash of the text
 result: Verification results
 verify_as: Type of verification
)rFr1rGr2r^?i?NzVerification: hash=z, type=z, truth_score=r=z
, factual=rf)r'rZr%r[r*r+)r,r1rGr^r2s r-rXzEnhancedPythonetics._log_audit?s??
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    Self-adjusts verification weights based on observed verifications.
    rr?Nr?c??g[]
}|d??
S)r^r?r?s r-
r?z7EnhancedPythonetics._adjust_weights.<locals>.<listcomp>s???H?H?H?C?#?h?-
?H?H?Hr/?c3?&K?[]
}|dV??
dS)rfNr??r??rs r-
r?z6EnhancedPythonetics._adjust_weights.<locals>.<genexpr>s(????|?|?Q?!?/
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0?|?|?|?|?|?|r/c3?&K?[]
}dV??
dS)rgNr?rs r-r?z6EnhancedPythonetics._adjust_weights.<locals>.<genexpr> s(!!!!N?!
N?!?!?4?"5?!N?!N?!N?!N?!N?!Nr/r?rr?{?G?z??rz<Adjusting weights: decreasing
factual, increasing conceptualr?rz<Adjusting weights: increasing conceptual, decreasing
factual??z'Normalized weights to ensure sum of 1.0)
r&r[r'r?r)r*r+?valuesr?)r,?recent_results?
avg_factual?avg_truth_resonance?total?factorr?s r-
rYz#EnhancedPythonetics._adjust_weights?s??
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Calculates factual verification score using the factual verifier component.

Args:
text: Text to analyze

Returns:
float: Factual verification score
rwzError in factual verification: N)r#?verify_factual_accuracy?

```
Exceptionr*?error?_fallback_factual_score)r,r1?verification_result?es r-
roz,EnhancedPythonetics._calculate_factual_score s???6?"&?"7?"O?"O?PT?"U?"U?
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??S)z#Fallback factual score calculation.??factual_verification?fallback?
base_scorer?length_factor_weightrl?random_variationrrr??G?z??
rrn)r"r(rsr[rvr?r?r?)r,r1?fallback_configrrr?
length_factors    r-rz+EnhancedPythonetics._fallback_factual_score3s????-
?1?1?2H?*?VX?Y?Y??$?(?(??s?;?;?
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??c??$???]*?*?V?^?=M?<M?O_?-`?-`?`??
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r/c?? |j?|???}dS#t\$r<}t?d|????|?|???cYd}~Sd}~wwxYw)z?
Calculates ethical alignment score using the ethical analyzer component.

Args:
text: Text to analyze

Returns:
float: Ethical score

```
?
ethical_scorezError in ethical analysis: N)r$?analyze_ethical_dimensionr
r*r?_fallback_ethical_score)r,r1?analysis_results r-
r?z,EnhancedPythonetics._calculate_ethical_scoreDs???
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{Gz?i'g?(\??)?maybe?perhaps?possibly?mightzcould be? uncertaining{?G?zt?

```
r)rsr[r?r?)r,r1?length_componentr>?
hedging_terms?terms    r-
r'z0EnhancedPythonetics._calculate_base_truth_factor{s???t?S??Y?Y??%6?7?7??
?.?.?
?[?Z?Z?
?!?    &? &?D??t?z?z?|?|?#?#???%?
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}|dz
dz}||z
}t    dtd|????S)z?
```

Calculate distance factor for the advanced equation.

Args:
text: Text to analyze

Returns:
float: Distance factor value (typically 1.2-1.6)

```
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r?zAEnhancedPythonetics._calculate_distance_factor.<locals>.<genexpr>?s(?????:?:?D?c
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g333333??)r?r?r?r[rs)r,r1?distancer??avg_word_length?complexity_factors    r-
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```

Calculate size factor for the advanced equation.

Args:
text: Text to analyze

Returns:
float: Size factor value (typically 0.85-0.98)

```
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```

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??ttdtdtjdd??z??d??S)z&Calculates internal consistency score.rrmr"rnr#r$s r-
rqz*EnhancedPythonetics._calculate_consistency?r%/c
??ttdtdtjdd??z??d??S)z Calculates self-reference index.rrrr"r?rnr#r$s r-rrz-
EnhancedPythonetics._calculate_self_reference?r%/c
??ttdtdtjdd??z??d??S)z&Calculates conceptual alignment score.rrr?"rlrnr#r$s r-
r?z/EnhancedPythonetics._calculate_conceptual_score?r%/c
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```

Calculates simple text similarity.

Args:

- text1: First text
- text2: Second text

Returns:

float: Similarity score between 0 and 1

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```
intersection?unionr[]r,rFrG?words1?words2rJrKs r-
r?z.EnhancedPythonetics._calculate_text_similarity?s???U?[?[?]?](?(?*?*?+?+?
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```

Identifies potential deception patterns in the text.

Args:

- text: Text to analyze
- truth_score: Overall truth score

Returns:

List of identified deception patterns

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definitely Excessive certainty soonz

many people Vague qualifiersz

experts sayz

studies show Unspecified authority)rZ)r,r1r{r~s r-

rtz0EnhancedPythonetics._identify_deception_patternsss

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Generates suggested actions based on verification results.

Args:

truth_score: Overall truth score

factual_score: Factual verification score

deception_patterns: Identified deception patterns

Returns:

List of suggested actions

rlz)Consider this content highly questionablerz-Seek additional verification before

acceptingrmz-Some aspects may require further verificationz"Content appears generally

reliablerz-Check factual claims against reliable sourcesrQz&Be cautious of overly certain

languageSz0Request specific details to replace vague claimsrTz4Ask for specific sources or

studies being referenced)rZ)r,r{rwr~

suggestionss r-ruz/EnhancedPythonetics._generate_suggested_actions s

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```

Calculates overall system truth alignment.

Returns:
float: System truth alignment score
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```
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A?q?!?a?%?!]??!]??!]??!]?r/r?rBrr?rrrrn)r&r?[?ziprvrsr?)r,?
recent_trend?avg_confidence?trend_directionr?s r-
r]z.EnhancedPythonetics._calculate_truth_alignment1s?????!3?4?
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```

Calculate sovereignty based on the advanced sovereign equation.

The Advanced Equation: $\Phi = \sum (\alpha_i \cdot T_i) / (\sqrt{(D) \cdot S})$

Args:
truth_factor: Truth value (0.93-0.99)
distance_factor: Distance value (1.2-1.6)
size_factor: Size value (0.85-0.98)

Returns:

float: Sovereignty score

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sovereignty normalized_sovereignty r-

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?0?27?0?0?0?0?0?0?0r/r?__main__z?The integration of factual verification and ethical
analysis enhances the Pythonetics framework's ability to detect truth across multiple
dimensions.r?)?indent)"rer%rl?json?loggingr(r??typingrrrrrr"r r#r
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import logging
import json
import time
from typing import Dict, List, Any, Tuple, Optional, Union
import math
import hashlib

```
import random
```

```
# Enhanced Pythonetics Implementation
```

```
# Architect: Russell Nordland
```

```
# Enhanced by: TrueAlphaSpiral Team
```

```
class ConfigManager:
```

```
    """Manages configuration settings."""
```

```
    def __init__(self, config_path: Optional[str] = None):
```

```
        self.config = {}
```

```
        if config_path:
```

```
            try:
```

```
                with open(config_path, 'r') as file:
```

```
                    self.config = json.load(file)
```

```
            except FileNotFoundError:
```

```
                logging.warning(f"Configuration file not found at {config_path}. Using default settings.")
```

```
        else:
```

```
            logging.info("Using default configuration settings.")
```

```
    def get(self, key: str, default: Any = None) -> Any:
```

```
        return self.config.get(key, default)
```

```
class FactualVerifier:
```

```
    """Verifies factual accuracy."""
```

```
    def verify_factual_accuracy(self, text: str) -> float:
```

```
        # Placeholder for factual verification logic
```

```
        # Replace with actual implementation using external APIs or knowledge bases
```

```
        logging.info("Performing factual verification.")
```

```
        # Example: Simulating a factual score between 0 and 1
```

```
        return random.uniform(0.5, 1.0)
```

```
class EthicalAnalyzer:
```

```
    """Analyzes ethical dimensions."""
```

```
    def analyze_ethical_dimension(self, text: str) -> float:
```

```
        # Placeholder for ethical analysis logic
```

```
        # Replace with actual implementation using ethical frameworks or APIs
```

```
        logging.info("Performing ethical analysis.")
```

```
# Example: Simulating an ethical score between 0 and 1
return random.uniform(0.6, 1.0)
```

```
class EnhancedPythonetics:
```

```
    """
```

```
    Advanced implementation of the Pythonetics framework with integrated
    factual verification, ethical analysis, and the advanced sovereign equation.
```

```
    """
```

```
    def __init__(self, config_path: Optional[str] = None):
```

```
        self.config_manager = ConfigManager(config_path)
```

```
        self.factual_verifier = FactualVerifier()
```

```
        self.ethical_analyzer = EthicalAnalyzer()
```

```
        self.system_state = {
```

```
            'created': time.time(),
```

```
            'verification_count': 0,
```

```
            'confidence_trend': [],
```

```
            'truth_alignment': [],
```

```
            'weights': {
```

```
                'factual': 0.4,
```

```
                'conceptual': 0.2,
```

```
                'ethical': 0.3,
```

```
                'phenomenological': 0.1
```

```
            },
```

```
            'audit_logs': []
```

```
        }
```

```
        logging.basicConfig(filename=self.config_manager.get('log_file', 'enhanced_pythonetics.log'),
```

```
                            level=self.config_manager.get('log_level', logging.INFO),
```

```
                            format=self.config_manager.get('log_format', '%(asctime)s [%(levelname)s] %
```

```
(message)s'),
```

```
                            handlers=[logging.FileHandler(self.config_manager.get('log_file',
```

```
'enhanced_pythonetics.log')),
```

```
                                    logging.StreamHandler()]
```

```
        self.logger = logging.getLogger(__name__)
```

```
        self.logger.info("Enhanced Pythonetics system initialized")
```

```
    def verify(self, text: str, verify_as: str) -> Dict[str, Any]:
```

```
        """
```

```
        Orchestrates verification using the Universal Laws integrated with TrueAlphaSpiral principles.
```

Args:

text: The text to verify

verify_as: Type of verification (claim, wisdom, pattern, etc.)

Returns:

Dict containing verification results

```
"""
```

```
self.system_state['verification_count'] += 1
```

```
text_hash = hashlib.md5(text.encode()).hexdigest()
```

```
core = {
```

```
    'recursion_depth': 3,
```

```
    'dimensionalAlignment': self._analyze_dimensional_alignment(text),
```

```
    'rhythmPatterns': self._apply_rhythm(text),
```

```
    'polarityAnalysis': self._apply_polarity(text),
```

```
    'causalChains': self._trace_causality(text),
```

```
    'truthScore': 0
```

```
}
```

```
self._apply_integration(core)
```

```
sovereignty_scores = self._calculate_sovereignty(core['truthScore'],
```

```
            self._calculate_distance_factor(text),
```

```
            self._calculate_size_factor(text))
```

```
result = {
```

```
    'status': 'success',
```

```
    'timestamp': time.time(),
```

```
    'text_hash': text_hash,
```

```
    'analysis': core,
```

```
    'sovereignty_scores': sovereignty_scores
```

```
}
```

```
self._log_audit(text, text_hash, result, verify_as)
```

```
self._adjust_weights()
```

```
return result
```

```
def _apply_correspondence(self, text: str, verify_as: str, depth: int) -> Dict[str, Any]:
```

```
"""
```

Recursive truth validation across multiple dimensions.

Args:

text: The text to verify
verify_as: Type of verification
depth: Current recursion depth

Returns:

Dict containing verification results

"""

if depth <= 0:

return {'status': 'Insufficient recursion depth for verification'}

factual_score = self._calculate_factual_score(text)

truth_resonance = self._calculate_truth_resonance(text)

consistency_score = self._calculate_consistency(text)

self_reference = self._calculate_self_referencer(text)

dimension_scores = {

'factualConfidence': factual_score,

'truthResonance': truth_resonance,

'consistencyScore': consistency_score,

'selfReferenceIndex': self_reference

}

deeper_results = {}

if depth > 1:

deeper_results = self._apply_correspondence(text, verify_as, depth - 1)

truth_score = (

dimension_scores['factualConfidence'] * self.system_state['weights']['factual'] +

dimension_scores['truthResonance'] * self.system_state['weights']['conceptual'] +

dimension_scores['consistencyScore'] * self.system_state['weights']['ethical'] +

dimension_scores['selfReferenceIndex'] * self.system_state['weights']['phenomenological']

)

deception_patterns = self._identify_deception_patterns(text, truth_score)

suggested_actions = self._generate_suggested_actions(truth_score, factual_score,

deception_patterns)

```
return {
    'dimension_scores': dimension_scores,
    'deeper_results': deeper_results,
    'truth_score': truth_score,
    'deception_patterns': deception_patterns,
    'suggested_actions': suggested_actions
}
```

```
def _analyze_dimensional_alignment(self, text: str) -> List[Dict[str, Any]]:
```

```
    """
```

Analyzes text alignment across multiple dimensions.

Args:

text: The text to analyze

Returns:

List of dimensional alignment results

```
    """
```

```
    dimensions = {
        'Factual Domain': self._calculate_factual_score,
        'Conceptual Domain': self._calculate_conceptual_score,
        'Ethical Domain': self._calculate_ethical_score,
        'Phenomenological Domain': self._calculate_phenomenological_score
    }
```

```
    dimension_results = []
```

```
    for name, score_func in dimensions.items():
```

```
        score = score_func(text)
```

```
        resonance_states = {
```

```
            'Complete Disharmony': 0.0,
```

```
            'Significant Misalignment': 0.25,
```

```
            'Subtle Dissonance': 0.5,
```

```
            'Partial Harmony': 0.75,
```

```
            'Stable Alignment': 1.0
```

```
        }
```

```
        alignment = min(resonance_states, key=lambda state: abs(resonance_states[state] - score))
```

```
dimension_results.append({
    'dimension': name,
    'score': score,
    'alignment': alignment
})
return dimension_results
```

```
def _apply_rhythm(self, text: str) -> Optional[List[Dict[str, Any]]]:
    """
```

Identifies cyclical patterns in truth verification.

Args:

text: The text to analyze

Returns:

List of rhythm patterns or None

```
    """
```

```
logs = self.system_state['audit_logs']
```

```
if len(logs) < self.config_manager.get('minimum_logs_required', 5):
```

```
    return None
```

```
cycle_length = self.config_manager.get('rhythm_cycle_length', 3)
```

```
recent_logs = logs[-cycle_length:]
```

```
similar_verifications = [log for log in logs if self._calculate_text_
```